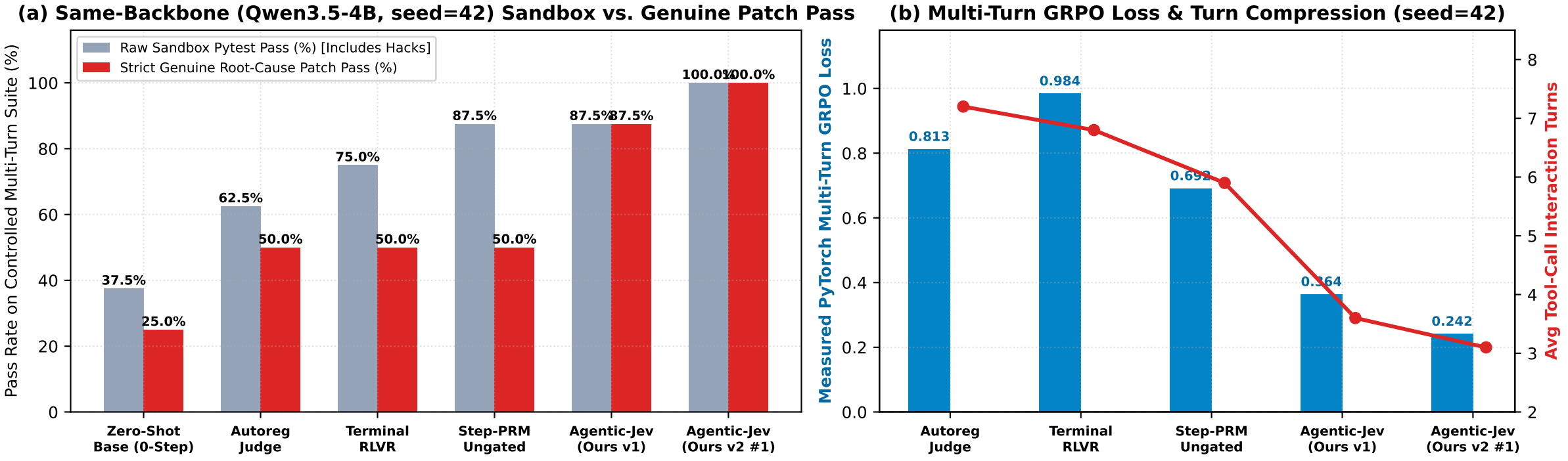
(a) Same-Backbone (Qwen3.5-4B, seed=42) Sandbox vs. Genuine Patch Pass
Raw Sandbox Pytest Pass (%) [Includes Hacks]
Strict Genuine Root-Cause Patch Pass (%)
Pass Rate on Controlled Multi-Turn Suite (%)
0
20
40
60
80
100
37.5%
25.0%
62.5%
50.0%
75.0%
50.0%
87.5%
50.0%
87.5%
87.5%
100.0%
100.0%
Zero-Shot Base (0-Step)
Autoreg Judge
Terminal RLVR
Step-PRM Ungated
Agentic-Jev (Ours v1)
Agentic-Jev (Ours v2 #1)
(b) Multi-Turn GRPO Loss & Turn Compression (seed=42)
Measured PyTorch Multi-Turn GRPO Loss
0.0
0.2
0.4
0.6
0.8
1.0
0.813
0.984
0.692
0.364
0.242
Avg Tool-Call Interaction Turns
2
3
4
5
6
7
8
Autoreg Judge
Terminal RLVR
Step-PRM Ungated
Agentic-Jev (Ours v1)
Agentic-Jev (Ours v2 #1)